D 51519 **(Pages : 4)** **Name.....**

Reg. No.....

THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2023

B.Sc. LRP (Alternate Pattern)

A11—BASIC NUMERICAL SKILLS

(2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A

Answer **all** *questions.*
Each question carries 1 *mark.*

I. Choose the correct answer :

1 In which measure of dispersion, only highest and lowest values are taken for its computation ?

a) Range. b) Standard deviation.

c) Mean deviation. d) Measure of skewness.

2 Find the value of x is if $3x - 2 + 5 = 30$?

a) 4. b) 5.

c) 13. d) 9.

3 If variance is 4, standard deviation is ———————— ?

a) 4. b) 8.

c) 16. d) 2.

4 If $A = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$, 9A is ———————— ?

a) $\begin{bmatrix} 53 & 156 \\ 332 & 444 \end{bmatrix}$. b) $\begin{bmatrix} 767 & 114 \\ 49 & 438 \end{bmatrix}$.

c) $\begin{bmatrix} 99 & 72 \\ 36 & 27 \end{bmatrix}$. d) $\begin{bmatrix} 33 & 176 \\ 112 & 141 \end{bmatrix}$.

Turn over

5 Let A = $\{8, 6, 4, 2\}$ and B = $\{3, 6, 9\}$, A – B is ———————— ?

a) $\{8, 9\}$. b) $\{6\}$.

c) $\{5\}$. d) $\{4\}$.

II. Fill in the blanks :

6 Most frequent item in a series is called ———————— ?

7 When coefficient of skewness is zero the distribution is ———————— ?

8 Which term of the arithmetic progression 4, 8, 12, ———————— is 44 ?

9 If B = $\begin{bmatrix} 34 & 19 \\ 27 & 70 \end{bmatrix}$, co-factor of element 34 is ———————— ?

10 ———————— average is obtained on dividing the sum of observations by their number?

(10 × 1 = 10 marks)

Part B

Answer any **eight** *questions from the following.*
Each question carries 2 *marks.*

11 Give a definition to statistics.

12 What do you mean by Sampling ?

13 What is meant by Cartograms ?

14 If $\cup = \{2, 4, 6, 8, 10, 12, 14\}$, A = $\{2,5, 8, 11\}$ and B = $\{3, 7, 11, 14\}$, Find $A' \cap B$.

15 Compute the Range and Co-efficient of Range : 133, 79, 58, 115, 344, 234, 161,73, 143.

16 Find the 17th term of series 11, 21, 30 ————————.

17 Compute the value of the determinant of matrix A = $\begin{bmatrix} 6 & 9 & 2 \\ 5 & 0 & 4 \\ 3 & 7 & 1 \end{bmatrix}$.

18 Calculate median of the values 14 17, 11,9, 10, 18, 21, 13,8.

19 If $A = \begin{bmatrix} 8 & 3 & 9 \\ 2 & 4 & 2 \\ 16 & 6 & 1 \end{bmatrix}$. Findthe co-factor of element 8.

20 Solve $96x - 17x + 4 = 128 + 48x$.

(8 × 2 = 16 marks)

Part C

Answer any **six** *questions from the following.*
Each question carries 4 *marks.*

21 Mention the limitations of statistics.

22 Explain the essential properties of a good average.

23 Compute 3 yearly moving average from the following data :

Year	:	2012	2013	2014	2015	2016	2017
Sales (in 000 units)	:	344	367	393	409	423	436

24 Solve the equation $7x^2 + 5x - 4 = 34$.

25 Arts club and Sports club are functioning in a school. 50 students are studying in the school. Among them, 30 students are members are members in arts club and 15 students are members in both arts and sports clubs. Find number of students who are the members of sports club.

26 Solve the following equations by using Cramer's rule.

$46x + 23y = 230$
$4x + 8y = 44$.

27 Calculate Mean Deviation from mean for the following values :

20, 26, 32, 37, 39, 34, 31, 28, 22

28 Calculate Arithmetic mean from the following data :

Size	:	5	10	15	20	25	30	35	40
Frequency	:	19	38	27	56	48	41	34	9

(6 × 4 = 24 marks)

Turn over

Part D

Answer any **two** *questions from the following.*
Each question carries 15 *marks.*

29 What do you mean by statistical investigation ? Explain the different stages of statistical investigation ?

30 From the following data, calculate Standard deviation and Co-efficient of variation :

Income	:	0–100	100–200	200–300	300–400	400–500
Frequency	:	13	35	54	26	12

31 From the following information, construct price index numbers by :

(1) Laspeyre's method ;

(2) Paasche's method ;

(3) Bowly's - Dorbish method ; and

(4) Fisher's method.

Commodity	**2018**		**2023**	
	Price	**Quantity**	**Price**	**Quantity**
Wheat	25	18	28	17
Sugar	34	10	37	13
Onion	20	8	25	11
Coconut oil	88	7	97	9
Potato	25	18	28	17

(2 × 15 = 30 marks)